Powerful Values and Local Obstructions in Cyclic Decimal Concatenations

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Abstract

We prove a quantitative density-zero bound for powerful values among cyclic concatenations of initial positive integers, uniformly in the rotation and in every fixed integer base. At the decimal boundaries $m = 10^d - 1$, we determine exact valuations and unit obstructions at the primes 3, 37, and 333667. The resulting congruences exclude every rotation for a set of digit lengths of lower density at least 0.85919.

1 Statements

In base $g \geq 2$, let W_m concatenate $1, 2, \ldots, m$, and let $D(m)$ be its digit length, with $W_0 = D(0) = 0$. For $0 \leq k < m$, moving the first k whole integer blocks to the end gives

$$R_{m,k} = g^{D(k)} W_m - (g^{D(m)} - 1) W_k. \tag{1}$$

In base ten these numbers, in increasing order, form OEIS A001292 [1]. Kashihara conjectured that none except 1 is a perfect power. Ripà [2] excludes $m \equiv 2, 3, 5, 6 \pmod 9$ by a digit-sum argument. The full conjecture remains open. We give a uniform sparsity bound and local obstructions for complete digit boundaries.

Recall that a positive integer is *powerful* if every prime divisor has exponent at least two. Every nontrivial perfect power is powerful. Logarithms in the following theorem and in Section 4 are natural.

Theorem 1 (Quantitative sparsity)**.** *For every fixed integer base $g \geq 2$,*

$$\#\{(m,k) : 2 \leq m \leq X,\ 0 \leq k < m,\ R_{m,k} \text{ powerful}\} \ll_g \frac{X^2 (\log\log X)^2}{\log X}. \tag{2}$$

For A001292, the corresponding count among the first N terms is $O(N(\log\log N)^2/\log N)$.

For the next result the base is ten and v_3 denotes the 3-adic valuation.

Theorem 2 (Complete digit boundaries)**.** *Let $d \geq 1$, $m = 10^d - 1$, $h = v_3(d)$, and $u = d/3^h$. Every rotation satisfies*

$$v_3(R_{m,k}) = h + 2, \qquad \frac{R_{m,k}}{3^{h+2}} \equiv u(2 + 6d) \pmod 9. \tag{3}$$

Corollary 1. *Put $v = h + 2$ and let $\mathcal{P}(v)$ be its set of prime divisors. No rotation at $m = 10^d - 1$ is a perfect power in any of the following cases:*

$\mathcal{P}(v)$	*condition on* u
$\{2\}$	$u \equiv 1 \pmod 3$
$\{3\}$	$u \equiv 1, 2, 7, 8 \pmod 9$
$\{2, 3\}$	$u \equiv 1, 7 \pmod 9$.

This includes $d \equiv 3, 6, 9, 21, 24 \pmod{27}$ and $d = 81$.

Proof. A prime exponent of a perfect power must divide v. A square requires v even and its normalized unit congruent to 1 modulo 3; by (3) this requires $u \equiv 2 \pmod 3$. A cube requires $3 \mid v$, hence $h \geq 1$ and $3 \mid d$. Its normalized unit must be $\pm 1 \pmod 9$, so $2u \equiv \pm 1 \pmod 9$, or $u \equiv 4, 5 \pmod 9$. The table excludes every available prime exponent. □

For the additional boundary primes, put

$$(p, s) \in \{(37, 3), (333667, 9)\}, \qquad C_s = \frac{9(11 - 10s)}{200} \pmod p.$$

Define

$$A_0 = 0, \qquad A_r = -9 \sum_{j=1}^{r} \frac{10^{j-1}}{10^j - 1} \pmod p \quad (0 < r < s).$$

All denominators here are units modulo p; $C_3 = 33$ and $C_9 = 221885$.

Theorem 3 (Two additional local obstructions)**.** *For every $d \geq 1$, write $d = sn + r$, $0 \leq r < s$. Then*

$$V_d := W_{10^d - 1} \equiv C_s n + A_r \pmod p, \qquad R_{10^d - 1, k} \equiv 10^{D(k)} V_d \pmod p. \tag{4}$$

If $s \mid d$, put $a = v_p(d)$ and $t = d/(sp^a)$. Every rotation has

$$v_p(R_{10^d - 1, k}) = a, \qquad \frac{R_{10^d - 1, k}}{p^a} \equiv 10^{D(k)} C_s t \pmod p. \tag{5}$$

Consequently, if such a rotation is an ℓth power for a prime ℓ, then $\ell \mid a$. If also $\ell \mid (p - 1)/s$, the unit $C_s t$ must be an ℓth power modulo p.

In particular, all rotations are excluded whenever $d = 111t$ with $37 \nmid t$, or $d = 3003003t$ with $333667 \nmid t$: their valuations at the respective prime are one. Combining known valuations also excludes any $3 \mid d$ with $\gcd(v_3(d) + 2, v_{37}(d)) = 1$.

Corollary 2 (Density of excluded complete boundaries)**.** *The set of positive integers d for which every rotation at $m = 10^d - 1$ is not a nontrivial perfect power has lower natural density at least*

$$\frac{286397396}{333333333} = 0.8591921888\ldots.$$

For example, all rotations are excluded at $d = 2, 12, 15, 18$.

2 Exact 3-adic values at complete boundaries

Write $V_d = W_{10^d - 1}$ and $S(d) = D(10^d - 1)$. We will prove the stronger intermediate congruence

$$\frac{V_d}{9d} \in \mathbb{Z}_3, \qquad \frac{V_d}{9d} \equiv 2 + 6d \pmod 9. \tag{6}$$

Quotients by d in this section are taken in $\mathbb{Q}_3$.

Lemma 1 (Averaging power series)**.** *For integers $d \geq 1$ and $n \geq 1$,*

$$v_3\left(\frac{1}{d} \sum_{j=1}^{d} j^n\right) \geq -\lfloor \log_3(n + 1) \rfloor. \tag{7}$$

Suppose $f(X) = \sum_{n \geq 0} c_n X^n$ has $v_3(c_n) \geq n$, $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. Then

$$\frac{1}{d} \sum_{j=1}^{d} f(j) \equiv 2 + 6d \pmod 9. \tag{8}$$

Proof. Using Stirling numbers of the second kind gives the exact identity

$$\frac{1}{d}\sum_{j=1}^{d} j^n = \sum_{r=0}^{n} \left\{ {n+1 \atop r+1} \right\} \frac{r!}{r+1}\binom{d-1}{r}.$$

Each denominator loses at most $\lfloor \log_3(n+1) \rfloor$ factors of 3, proving (7). For $n \geq 3$, $n - \lfloor \log_3(n+1) \rfloor \geq 2$, so all these terms vanish modulo 9 after averaging. The same holds for $n = 2$ by its stronger coefficient bound. Only $c_0 + c_1(d+1)/2$ remains, congruent to $5+3(d+1)/2 = 2+6d \pmod 9$. The coefficient bounds also justify convergence and termwise averaging. □

Define the 3-adic logarithm explicitly by

$$\ell = \sum_{r\geq 1} \frac{(-1)^{r+1} 9^r}{r}, \qquad v_3(\ell) = 2.$$

In the following power series put

$$E = e^{\ell X}, \quad a = E/10, \quad q = 9a, \quad b = E - 1, \quad S(X) = \frac{(9X-1)E+1}{9},$$

and define

$$\frac{U(X)}{9} = a^2 + \frac{ab+1}{9}\sum_{r\geq 0}\binom{q}{r+2} b^r, \qquad f(X) = \frac{U(X)}{9} e^{-\ell S(X)}. \tag{9}$$

At every positive integer j, $S(j)$ is the digit length of V_j, and $U(j)$ concatenates all j-digit integers. Indeed, with $a = 10^{j-1}$, $b = 10^j - 1$, and $q = 9a$, the usual arithmetic-geometric sum gives

$$b^2 U(j) = (ab+1)(1+b)^q - ((1+b)b+1).$$

Expanding $(1+b)^q$ cancels its constant and linear terms and gives (9), since $a + q = 10^j$ and $aq = 9a^2$.

Lemma 2. *The coefficients of f satisfy all hypotheses of Lemma 1.*

Proof. Let $\mathcal{A}$ consist of series whose degree-n coefficient has valuation at least n. It is closed under addition and multiplication. In the rth summand of $U/9$, the factor b^r makes all degrees below r zero. The factor q in $\binom{q}{r+2}$ cancels the displayed denominator 9. The remaining expression is a polynomial in E with 3-adic denominators bounded by $(r+2)!$. Its degree-n coefficient, for $r \leq n$, therefore has valuation at least

$$2n - v_3(n!) - v_3((r+2)!) \geq n.$$

Here $v_3(n!) \leq (n-1)/2$ for $n \geq 1$, and $v_3((r+2)!) \leq (n+1)/2$; the degree-zero case is immediate. The term a^2 satisfies the same bound. Thus $U/9 \in \mathcal{A}$.

The constant term of S is zero. Its linear coefficient is $1 - \ell/9$, of valuation at least 2, and for $n \geq 2$ its coefficient is $\ell^{n-1}(n - \ell/9)/n!$, of valuation at least n. Hence $S \in \mathcal{A}$. Expanding $e^{-\ell S}$ preserves these bounds: its rth exponential term gains $2r - v_3(r!) \geq 0$ factors of 3. Consequently $f \in \mathcal{A}$.

Expansion through degree two in (9) gives

$$\begin{aligned} c_0 &= \frac{1}{200}, \\ c_1 &= \frac{169\ell}{3000} + \frac{\ell^2}{1800}, \\ c_2 &= \frac{671\ell^2}{80000} + \frac{353\ell^3}{54000} + \frac{\ell^4}{32400}. \end{aligned}$$

Since $\ell/9 \equiv 1 \pmod 9$, these imply $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. □

Proof of Theorem 2. The concatenation of the complete digit blocks yields

$$V_d = 10^{S(d)} \sum_{j=1}^{d} U(j) 10^{-S(j)} = 9\, 10^{S(d)} \sum_{j=1}^{d} f(j).$$

Lemma 1 and $10^{S(d)} \equiv 1 \pmod 9$ prove (6). Its right side is a 3-adic unit, so $v_3(V_d) = h + 2$.

To transfer the normalized unit to every rotation, note that

$$S(d) = \frac{9d(d+1)}{2}\left(1 + \sum_{r=1}^{d-1} \frac{2 \cdot 9^r}{r+2}\binom{d-1}{r}\right). \tag{10}$$

This follows by expanding $S(d) = 9\sum_{j=1}^{d} j10^{j-1}$ in powers of 9. Each term after 1 in the bracket is divisible by 3; hence $v_3(S(d)) = 2 + h + v_3(d+1)$. Lifting the exponent gives

$$v_3(10^{S(d)} - 1) = 4 + h + v_3(d+1).$$

In (1), $10^{D(k)} \equiv 1 \pmod 9$ and the second term is divisible by 3^{h+4}. Thus $R_{m,k} \equiv V_d \pmod{3^{h+4}}$. Combining this with (6) proves both assertions in (3). □

3 The primes 37 and 333667

The proof first computes complete digit blocks modulo p, then averages their analytic interpolation to retain the exact valuation when $s \mid d$. For both pairs in Theorem 3,

$$\operatorname{ord}_p(10) = s \mid 9, \qquad v_p(10^s - 1) = 1, \qquad S(j) := D(10^j - 1) \in 9\mathbb{Z}.$$

Let U_j concatenate the j-digit integers, and put $B = 10^j$, $b = B - 1$, $a = B/10$, $q = 9a$. Its length jq is divisible by s. If $s \nmid j$, the block formula used in (9) gives $U_j \equiv -9a/b \pmod p$. If $s \mid j$, its multiplier B is one modulo p, so summing the appended integers gives $U_j \equiv 9/200 \pmod p$. Thus $U_j \pmod p$ has period s. Pairing each nonidentity sth root $z = 10^j$ with z^{-1} yields

$$\sum_{j=1}^{s-1} \frac{10^j}{10^j - 1} = \frac{s-1}{2}, \qquad \sum_{j=1}^{s} U_j \equiv -\frac{9(s-1)}{20} + \frac{9}{200} = C_s \pmod p.$$

All block multipliers are one modulo p, proving (4), including its rotation assertion.

To lift this calculation, set $Q = 10^s$, $L = \log Q \in p\mathbb{Z}_p$. For $r = 1, \dots, s$ and a variable $X \in \mathbb{Z}_p$, put

$$j = sX + r, \quad B = 10^r e^{LX}, \quad a = B/10, \quad q = 9a, \quad b = B - 1, \quad S_r = \frac{(9j-1)B + 1}{9}.$$

Define

$$U_r(X) = \frac{(ab+1)e^{L(X+r/s)q} - (Bb+1)}{b^2}, \qquad f(X) = \sum_{r=1}^{s} U_r(X)e^{-LS_r/s}.$$

At nonnegative integral X, these are the actual blocks and normalized blocks: q and $S(j)$ are divisible by s, which justifies the exponential expressions. For $r = s$, remove the apparent singularity by the convergent series

$$U_s(X) = aq + (ab+1)\sum_{t \ge 0} \binom{q}{t+2} b^t. \tag{11}$$

Here $b \in p\mathbb{Z}_p$ throughout $\mathbb{Z}_p$.

Write $f(X) = \sum_{n \ge 0} c_n X^n$. We claim

$$v_p(c_n) \ge n - v_p(n!) - \left\lfloor \frac{n+1}{p-1} \right\rfloor \ge \left\lceil \frac{p-3}{p-1} n \right\rceil \quad (n \ge 1). \tag{12}$$

Indeed, every nth derivative of a polynomial in e^{LX} is divisible by p^n. The exponents $L(X + r/s)q$ and $-LS_r/s$ have the same derivative property for $n \ge 1$; differentiating their exponentials preserves it. When $r < s$, b is a unit and its inverse also has this property. In the tth summand of (11), differentiating b^t gives at least $p^{\max(t,n)}$ after combining with the other factors: an undifferentiated factor b contributes p, and a factor differentiated $i \ge 1$ times contributes p^i. The binomial coefficient loses at most $v_p((t+2)!) \le \lfloor (t+1)/(p-1) \rfloor$. Thus the nth derivative of the whole sum has valuation at least $n - \lfloor (n+1)/(p-1) \rfloor$; this follows for $t \le n$ directly and for $t > n$ because $t - \lfloor (t+1)/(p-1) \rfloor$ is nondecreasing. The tails converge with all derivatives. Multiplication by $e^{-LS_r/s}$ preserves

this bound, and division by $n!$ proves the first inequality. The second uses $v_p(n!) \le (n-1)/(p-1)$ and integrality of valuations.

The Stirling-number argument for (7), with p in place of 3, also gives

$$v_p\left(\frac{1}{N}\sum_{i=0}^{N-1} i^n\right) \ge -\lfloor \log_p(n+1)\rfloor.$$

For $p \ge 7$ and $n \ge 1$, the final bound in (12) is at least $1 + \lfloor \log_p(n+1)\rfloor$. Hence termwise averaging gives, for every $N \ge 1$,

$$\frac{1}{N}\sum_{i=0}^{N-1} f(i) \equiv c_0 \equiv C_s \pmod p.$$

The concatenation identity $V_{sN} = 10^{S(sN)}\sum_{i=0}^{N-1} f(i)$ and $C_s \not\equiv 0 \pmod p$ show $v_p(V_{sN}) = v_p(N)$ and give its normalized unit in (5). For $s \mid d$, the identity

$$S(d) = d10^d - \frac{10^d - 1}{9}$$

and lifting the exponent give $v_p(S(d)) = v_p(d) = a$, followed by $v_p(10^{S(d)} - 1) = a + 1$. Applying (1) transfers the valuation and unit to every rotation. Since 10 has order s, it is an ℓth power modulo p whenever $\ell \mid (p-1)/s$. This proves all of Theorem 3.

Proof of Corollary 2. The 3-adic restrictions leave only

$$d \equiv 2 \pmod 3, \qquad d \equiv 12, 15 \pmod{27}, \qquad d \equiv 18 \pmod{27}, \qquad 27 \mid d.$$

In the first and third sets, a perfect power must be a square; in the second, a cube. In each fixed class modulo 27, the affine formula (4) makes V_d run uniformly modulo each p as d varies modulo $27p$. The two residues are independent by the Chinese remainder theorem, since $C_s \ne 0$ and $s \mid 9$. The proportions of residues that are zero or squares, or zero or cubes, at both primes are respectively

$$\alpha = \frac{19}{37}\frac{166834}{333667} = \frac{3169846}{12345679}, \qquad \beta = \frac{13}{37}\frac{111223}{333667} = \frac{1445899}{12345679}.$$

Multiplication by $10^{D(k)}$ preserves both power classes. Bounding the last set trivially, the upper density of the remaining d is at most $\alpha(1/3+1/27)+2\beta/27+1/27$. Subtracting from one proves the stated bound. For the examples, $V_2 \equiv 242665 \pmod{333667}$ is a nonsquare, $V_{12}, V_{15} \equiv 21, 17 \pmod{37}$ are noncubes, and $V_{18} \equiv 13 \pmod{37}$ is a nonsquare. □

4 A uniform sieve for powerful values

We now prove Theorem 1. In a fixed d-digit interval set

$$B = g^d, \quad L = g^{d-1}, \quad c = B - 1, \quad A = c^2 W_{L-1} + Lc + 1.$$

The recurrence $W_{m+1} = BW_m + (m+1)$ gives, for $L - 1 \le m \le B - 1$,

$$c^2 W_m = AB^{m-L+1} - ((m+1)c + 1), \qquad D(m) = D(L-1) + d(m - L + 1).$$

For fixed k, writing $u = g^{D(k)}$, substitution into (1) gives

$$R_{m,k} = \alpha B^m + \beta m + \gamma, \quad \begin{cases} \alpha = (uA/c^2 - g^{D(L-1)}W_k)B^{1-L}, \\ \beta = -u/c, \quad \gamma = W_k - uB/c^2. \end{cases} \tag{13}$$

At every prime $p \nmid g(B-1)$ the coefficients are defined modulo p^2 and β is a unit. The right side provides a periodic modular extension; only its values inside the digit interval are identified with the concatenation.

Lemma 3 (Valuation-one densities)**.** *Let $p \nmid B\beta$ and $F(n) = \alpha B^n + \beta n + \gamma \pmod{p^2}$. The event $p \mid F(n)$, $p^2 \nmid F(n)$ has period dividing $T_p = p^2(p-1)$ and density $\rho_p = (p-1)/p^2$. For any finite set P of eligible primes the events have joint density $\prod_{p\in P}\rho_p$ in a period dividing $\operatorname{lcm}_{p\in P} T_p$.*

Proof. Fix $n \pmod{p-1}$. The exponential term is constant modulo p, and the linear coefficient is nonzero, so exactly one class $n \pmod{p}$ is a root. Its p lifts differ by $p(p-1)t$. Euler's theorem fixes the exponential term modulo p^2 on these lifts, while the linear term takes p distinct values. Precisely one lift is zero modulo p^2, leaving $p-1$ of valuation one.

For the joint assertion induct on the largest prime q. The smaller-prime events have a period T with no prime divisor as large as q. Thus $q \nmid H = \operatorname{lcm}(T, q-1)$. In each fixed class modulo H, the smaller-prime events are fixed and the Chinese remainder theorem lets $n \pmod{q^2}$ vary freely. The local count gives the same fraction ρ_q in every class. Averaging multiplies the previous joint density by ρ_q. This allows overlapping periods, including a smaller prime dividing a larger prime minus one. □

For an interval I of H integers and a subset P', Lemma 3 gives its simultaneous event count as $H\prod_{p\in P'}\rho_p + O(\prod_{p\in P'}p^3)$, since the interval error is at most the joint period. Take eligible primes at most z, put $\lambda = \sum \rho_p$, and choose an even integer $K \geq 0$. Powerful values avoid every valuation-one event. Finite Bonferroni gives

$$\#\{n \in I : F(n) \text{ powerful}\} \leq H\left(e^{-\lambda} + \frac{\lambda^{K+1}}{(K+1)!}\right) + (K+1)z^{4K}. \tag{14}$$

Indeed, if e_j are the elementary symmetric sums of the ρ_p, the even truncated main term is at most $\prod(1-\rho_p)+e_{K+1} \leq e^{-\lambda} + \lambda^{K+1}/(K+1)!$. There are at most z^j subsets of size j, each costing at most z^{3j}, which proves the displayed error. These estimates are uniform in the coefficients of (13) and in k.

Choose

$$\eta = \frac{\log g}{100}, \qquad z = \exp\left(\frac{\eta d}{\log d}\right), \qquad K = 2\lceil 4\log d\rceil,$$

and use primes $d^2 < p \leq z$ with $p \nmid g(g^d-1)$. For large d, $z > d^2$. Omitted primes exceeding d^2 and dividing g^d-1 number at most $\log(g^d-1)/(2\log d)$, with reciprocal sum $O_g(1/(d\log d))$. The fixed prime divisors of g eventually lie below d^2. Mertens' estimate [3] therefore yields

$$\lambda = \log\log z - \log\log(d^2) + O_g(1) = \log d - 2\log\log d + O_g(1).$$

In particular $0 < \lambda \leq \log d$ eventually, so $e^{-\lambda} \ll_g (\log d)^2/d$. Also

$$\frac{\lambda^{K+1}}{(K+1)!} \leq (e/8)^{K+1} \leq d^{-8}, \qquad (K+1)z^{4K} < g^{d/2}$$

for large d; the latter follows from $K \leq 10\log d$ and $\log((K+1)z^{4K}) \leq \log(K+1) + 40\eta d < d\log g/2$. Thus for every fixed k and every interval inside the digit block with $m > k$,

$$\#\{m \in I : R_{m,k} \text{ powerful}\} \ll_g |I|\frac{(\log d)^2}{d} + g^{d/2}. \tag{15}$$

Let D be the digit length of $\lfloor X\rfloor$. For each $d \leq D$ and k, apply (15) on $\max(g^{d-1}, k+1, 2) \leq m \leq \min(X, g^d-1)$. There are at most g^d choices of k, so the total interval error is $\sum_{d\leq D} g^{3d/2} = O_g(X^{3/2})$. Blocks with $d \leq D/2$ contain only $O_g(X)$ pairs and may be bounded trivially. On the other blocks the main factor is $O((\log D)^2/D)$, and there are $O(X^2)$ pairs. This proves (2).

Finally, each m supplies m distinct rotations. If m has $d \geq 2$ digits, the cyclic digit word has a unique occurrence of 1 followed by $d-1$ zeros: it is the integer g^{d-1}. Such a zero run cannot cross an integer boundary or the cyclic join, whose next digit is 1. The unique marker prevents a nonidentity digit rotation from fixing the word. For $d = 1$ all digits are distinct. Digit lengths increase strictly with m, so the sequence through block $m = X$ has $X(X+1)/2$ entries. Bounding a partial final block by the whole block and taking $X \asymp \sqrt{N}$ proves the sequence-index assertion. □

References

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